

The 2019 classification put to the test, and the eight pairs

Alberto Acedo Biome Makers Inc. Draft v0.2, 8 September 2026. Supersedes v0.1, which used the 2010 classification.

Preliminary note

The previous version of this note tested Montecchi’s classification (2010, Kadmos 49). The author has pointed out that it was revised and superseded in her 2019 book, *Contare a Haghia Triada*, pp. 97-107, and has made the volume available. This version uses the revised classification and adds the test the new version makes possible.

1. The two classifications, measured with the same yardstick

classification	tablets	pairs, same series	pairs, different series	difference	null	p
2010, logograms	88	0.210	0.036	0.174	0.000	< 0.0005
2019, logograms	131	0.133	0.026	0.107	0.000	< 0.0005
2010, vocabulary	88	0.044	0.012	0.031	0.000	< 0.0005
2019, vocabulary	131	0.034	0.010	0.024	0.000	< 0.0005

Both hold with the same force. The 2019 classification shows somewhat lower internal cohesion (0.133 against 0.210 for logograms), and the reason is arithmetic rather than methodological: it classifies 131 tablets into 39 classes and series, against 88 into 23 in 2010. Distributing more documents across more groups makes the groups finer and more numerous, so the average within-group similarity necessarily falls. The ratio between inside and outside holds: 5.1 times in 2019, 5.8 in 2010.

That is, **the 2019 revision extends coverage of the corpus by 49% without losing discrimination**, which is exactly what is asked of a revised classification.

2. The test the 2019 classification makes possible: the eight pairs

The most testable novelty of the book is that the author marks **eight pairs of tablets as homogeneous** by subject, text structure, find-spot and, in most cases, scribal hand. That is a much finer claim than grouping into series, and it can be measured directly.

mark	tablets	Jaccard of logograms	Jaccard of vocabulary
[1]	HT 85 / HT 122	0.429	0.107
[2]	HT 93 / HT 102	0.125	0.118
[3]	HT 86 / HT 95	0.000	0.545
[4]	HT 17 / HT 19	1.000	0.333
[5]	HT 16 / HT 20	0.400	0.000
[6]	HT 32 / HT 33	0.067	0.500
[7]	HT 114 / HT 121	0.833	0.667
[8]	HT 87 / HT 117	0.000	0.071

Mean of the eight pairs: 0.357 for logograms and 0.293 for vocabulary. Mean of pairs from the same series: 0.133 and 0.034. Mean of all pairs in the corpus: 0.030 and 0.011.

The eight pairs are nearly three times as cohesive as any two tablets from the same series, and twelve times as cohesive as any two tablets in the corpus. Against a null drawing eight pairs at random from among same-series pairs, $p = 0.0038$; against eight pairs drawn at random from the corpus, $p < 0.0002$.

That is the strongest result in this note, and it is worth stating what it means: **the author identified by hand, with no similarity measure of any kind, the eight most cohesive pairs in a corpus of a hundred and thirty tablets.** Not the series, which are a judgement of content, but the specific pairs, which is a judgement of detail.

3. The three cases that show why both measures are needed

Three pairs are highly cohesive on one measure and null on the other, and each illustrates a different kind of kinship:

HT 86 / HT 95 (mark [3], series Ed): logograms 0.000, vocabulary 0.545, the highest figure in the corpus. They share no commodity and share more than half their words: both record KU-NI-SU, SA-RU, DI-DE-RU, DA-ME, MI-NU-TE and QE-RA₂-U with quantities of 10 and 20. They are two versions of the same document, and a logogram-based measure would not see them.

HT 17 / HT 19 (mark [4], series Ga): logograms 1.000, vocabulary 0.333. They share the whole of their commodities. This is kinship of type, not of content.

HT 114 / HT 121 (mark [7], series Ud): 0.833 and 0.667, high on both. Complete kinship, and they are also by the same hand (no. 4 in the author's table) and from the same room.

HT 87 / HT 117 (mark [8], series Va) gives 0.000 and 0.071, and is the only one of the eight that neither measure sees. The reason lies in the classification itself: class V is the one for tablets **without an ideogram**, so there are no logograms to compare, and their kinship is one of text

structure, which neither of our measures captures. It is a limitation of the instrument and not of the classification, and the same one already observed with the 2010 version.

4. Series, hand and room: the three factors separated

The 2019 table records the **room of discovery**, which the 2010 version did not. With it, three factors hitherto confounded can be separated for the first time: what the document is about (the series), who wrote it (the hand) and where it was kept (the room).

Of the 135 classified tablets, all have a room, 59 have a hand and 92 also have enough vocabulary to measure. There are eight rooms: Villa 59 (50 tablets), House of the Cauldron 7 (34), House of the Cauldron 9 (20), Villa 72 (10), and four smaller ones.

All three at once (46 tablets with all three data)

series	hand	room	pairs	shared vocabulary
same	same	same	5	0.207
same	different	different	11	0.094
same	same	different	4	0.045
different	same	same	28	0.028
same	different	same	12	0.021
different	different	different	688	0.018
different	different	same	262	0.017
different	same	different	25	0.016

The cell agreeing on all three dominates every other, at more than twice the next and eleven times the corpus mean. With five pairs it is an indication and not a measurement.

The two with sufficient sample (92 tablets)

series	room	pairs	shared vocabulary
same	same	53	0.0805
same	different	66	0.0432
different	same	974	0.0196
different	different	3,093	0.0148

With permutation within strata: **the effect of series within the same room is 0.0608 (null -0.002; $p < 0.002$); the effect of room within the same series is 0.0373 (null 0.024; $p = 0.212$, not significant).**

The reading

The series explains shared vocabulary; the room does not. Two tablets of the same series resemble each other even when kept in different rooms (0.043, three times the corpus mean); two tablets from the same room do not resemble each other unless they belong to the same series (0.020, barely above the mean).

This is what one would expect of an archive organised by subject and by responsible official, and not of a deposit arranged for storage convenience. And it has a consequence for the archaeological use of the datum: **find-spot alone is not sufficient to infer documentary kinship**, however accessible a criterion it is.

The full hierarchy of factors

factor	effect	significance
series and hand both agreeing	0.216	the dominant cell
series, hand and room all agreeing	0.207	indication, 5 pairs
hand within the same series	0.164	$p < 0.003$
series within the same room	0.061	$p < 0.002$
room within the same series	0.037	not significant

The vocabulary of a tablet is determined by the scribe working within a particular dossier; where it ended up stored adds nothing.

A caution about the rooms

In the 2019 table many room attributions carry a question mark, since they derive from the reconstruction of early twentieth-century excavations. They have been treated here as firm, because excluding them would leave the sample without size. The uncertainty runs in the conservative direction: were the rooms better determined, the effect might appear, not disappear. What is sustained is therefore that **with the provenance data available the room does not explain shared vocabulary**, and not that it would fail to do so with better data.

5. Reproducibility

All figures in this note are produced by two functions of the kuro package, `group_cohesion` (cohesion of a grouping against label permutation) and `factor_effects` (mean similarity per combination of agreeing factors), and have been verified by recomputing them with independent code and a different seed: the eight pairs give 0.3567 and 0.2927, same-series pairs 0.1333, and the four series-by-room cells agree to the fourth decimal. The 2019 classification transcribed from table 4 accompanies this work as a data file.

Data

Classification: Montecchi (2019), *Contare a Haghia Triada*, Incunabula Graeca CVII, table 4, pp. 97-104. Corpus: GORILA via the LinearA Explorer (Hogan 2022), with the GORILA V hands. Scripts: the kuro package.

Declaration of assistance

Corpus processing, the permutation tests and the draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

References

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